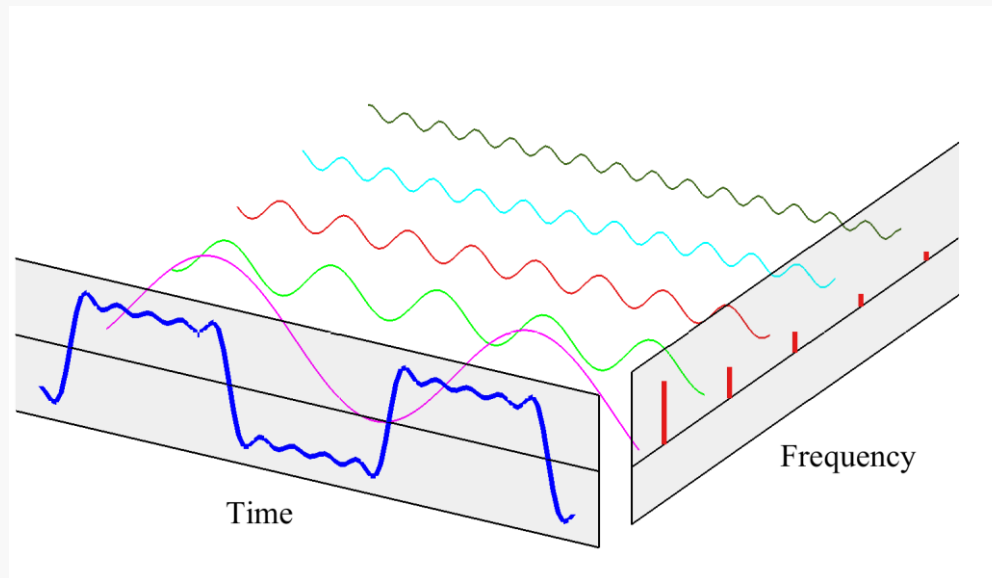


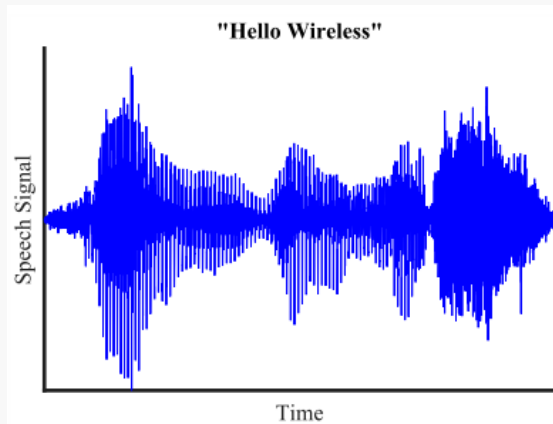
Introduction to Signals

Qasim Chaudhari



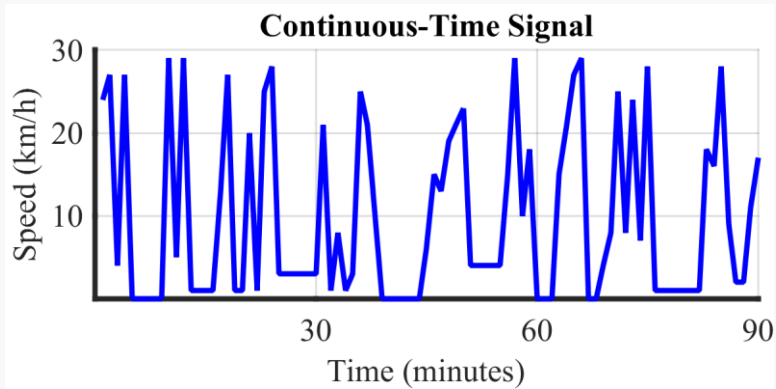
Signals

A signal is any measurable quantity that varies with time

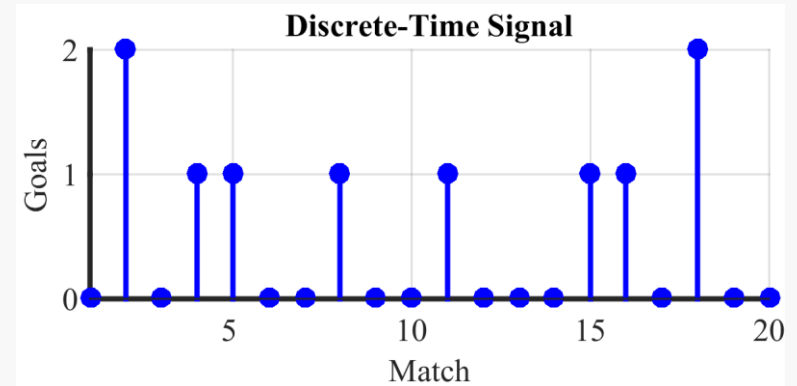


Continuous-Time vs Discrete-Time

Player speed

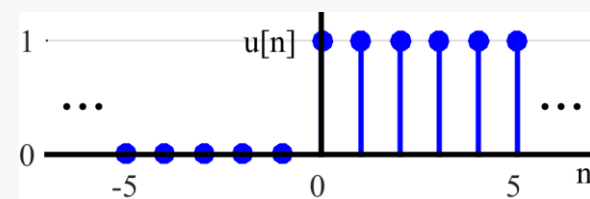
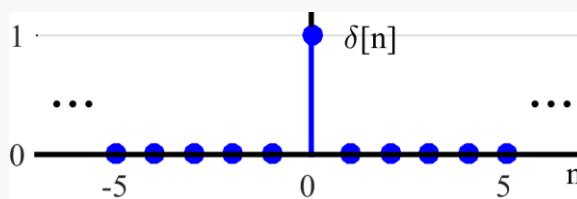


Player goals

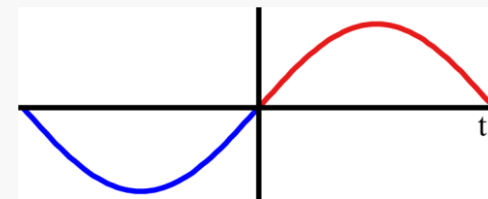
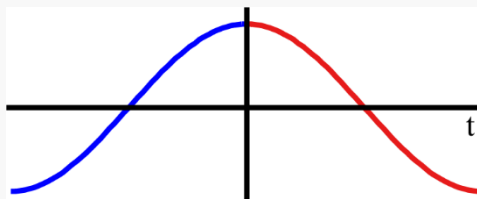


Basic Signals

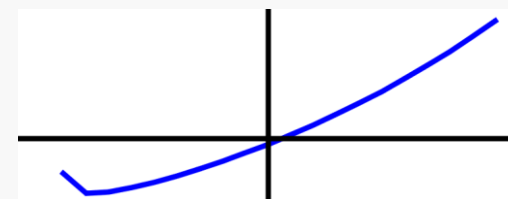
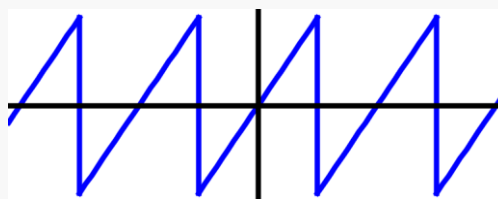
- Unit impulse and unit step



- Even and odd



- Periodic and aperiodic

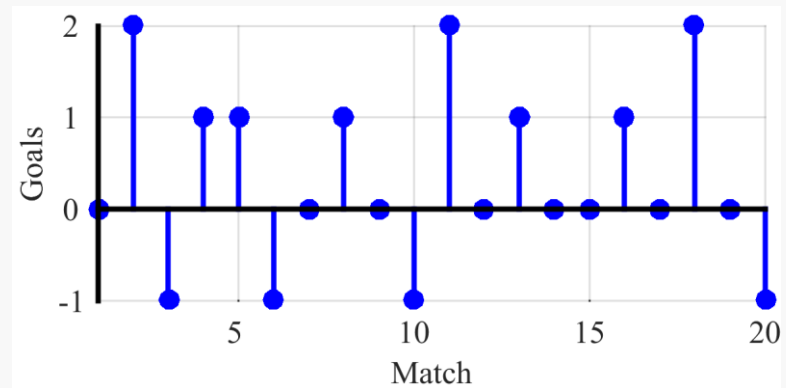


Signal Energy

A measure of signal strength or size

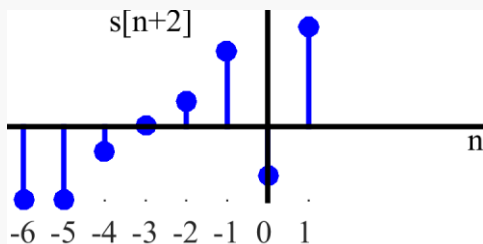
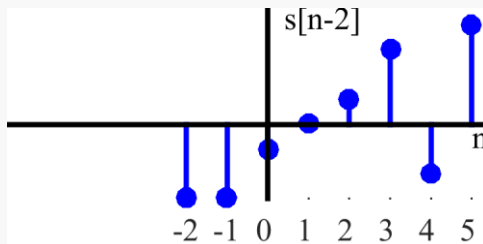
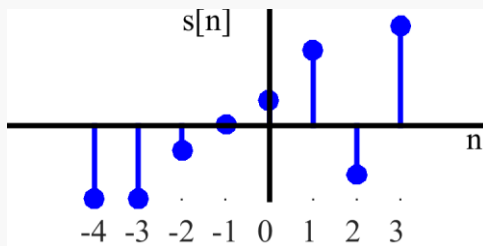
Both positive and negative values are present

$$E_s = \sum_n |s[n]|^2$$

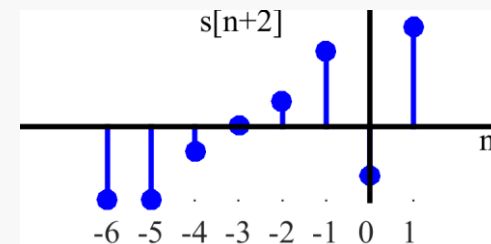
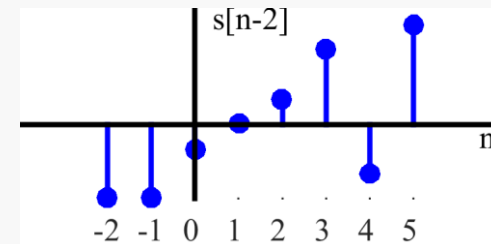
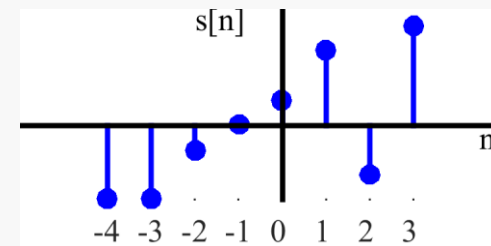


Time Shift

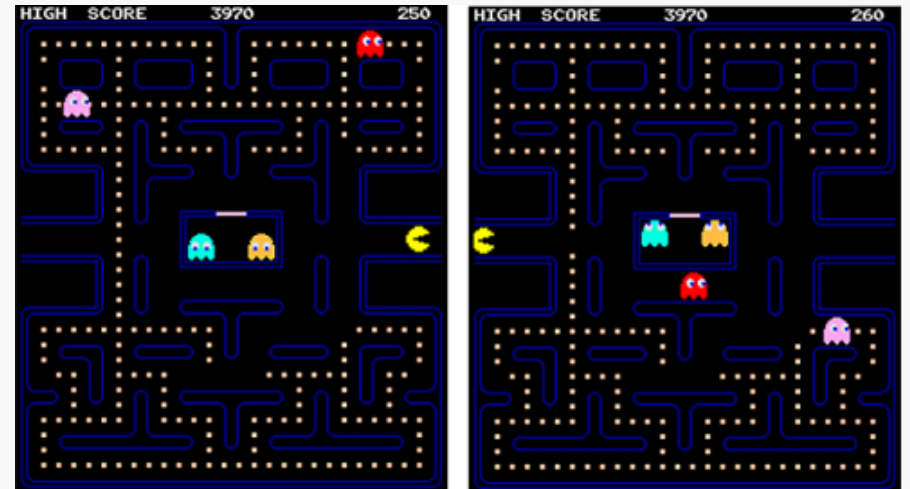
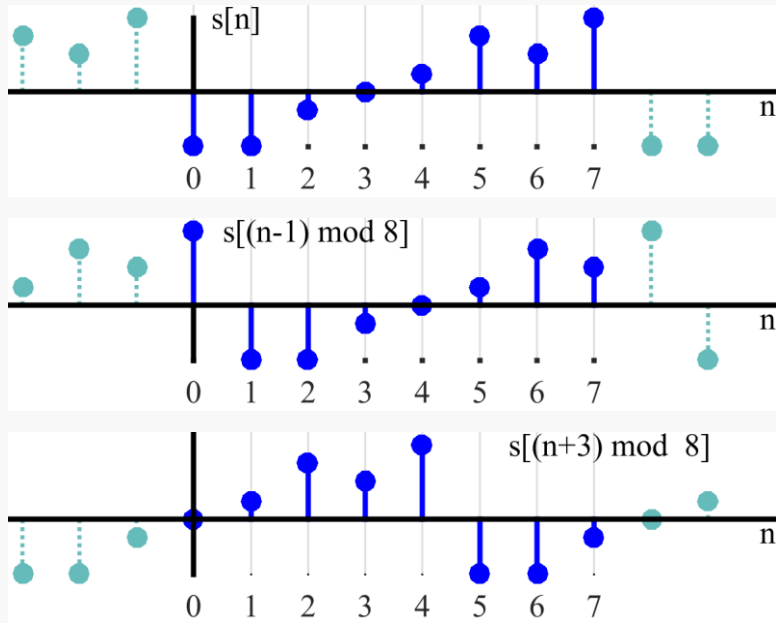
Shifting the signal



Shifting the axis



Circular Shift



Complex Numbers

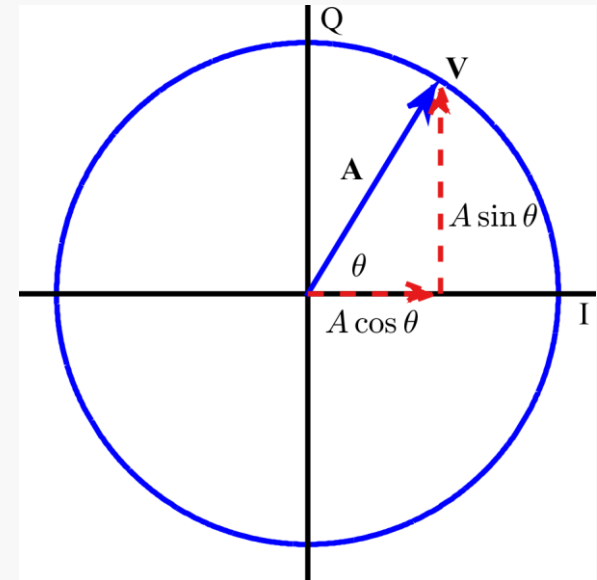
- An ordered pair of real numbers in (x,y) plane
- I and Q parts

$$V_I = |V| \cos \angle V$$
$$V_Q = |V| \sin \angle V$$

- Magnitude

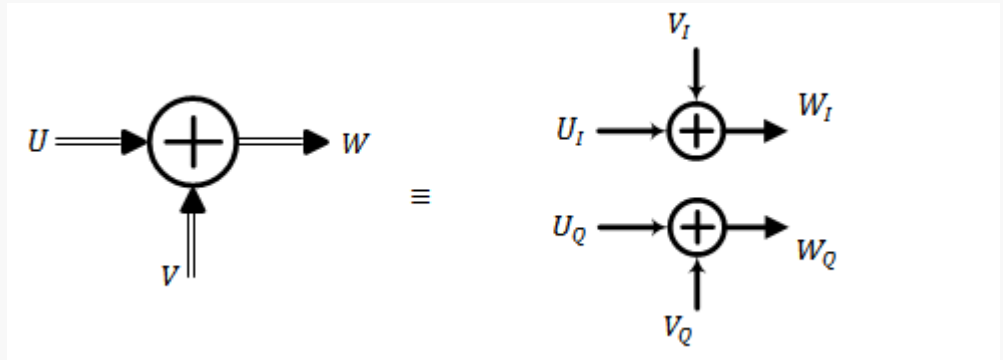
$$|V| = \sqrt{V_I^2 + V_Q^2}$$

- Phase?



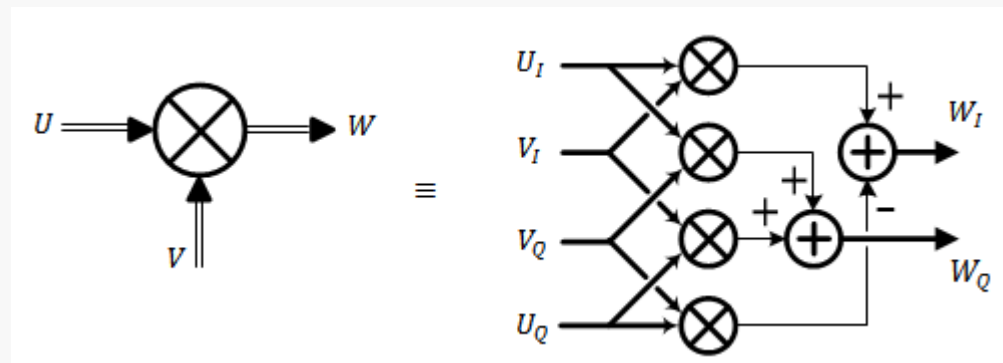
Addition and Multiplication

Simply add the respective I and Q parts



Multiplication rule of complex numbers

$$\begin{aligned} I &\rightarrow I \cdot I - Q \cdot Q \\ Q &\rightarrow Q \cdot I + I \cdot Q \end{aligned}$$



Complex Conjugate

Negate the Q component

$$\begin{aligned}\{V^*\}_I &= V_I \\ \{V^*\}_Q &= -V_Q\end{aligned}$$

Same as negating the phase

$$\begin{aligned}|V^*| &= |V| \\ \angle V^* &= -\angle V\end{aligned}$$

Correlation

Recall Sherlock Holmes

- The world is full of obvious things which nobody by any chance ever observes.

Holmes to Watson - The Hound of the Baskervilles

- ... and the marks of his teeth are very plainly visible. The dog's jaw, as shown in the space between these marks, is too broad in my opinion for a terrier and not broad enough for a mastiff. It may have been -- yes, by Jove, it is a curly-haired spaniel.

Sherlock Holmes - The Hound of the Baskervilles

Correlation

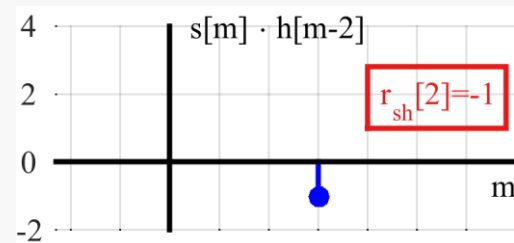
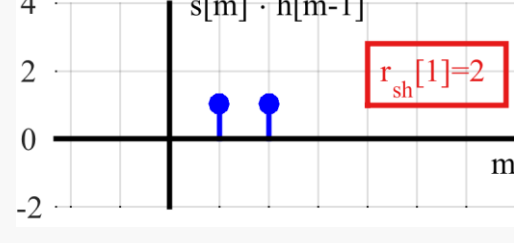
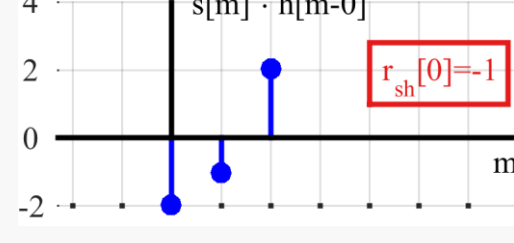
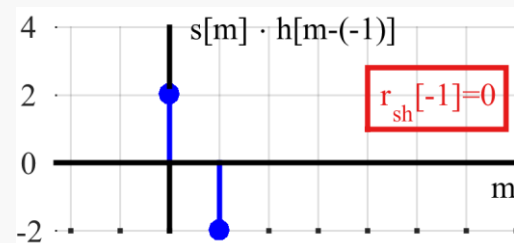
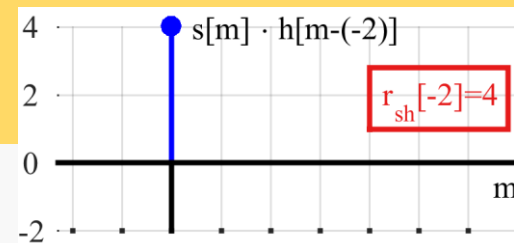
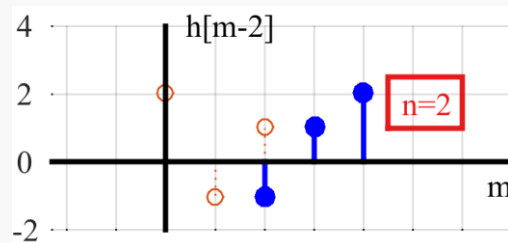
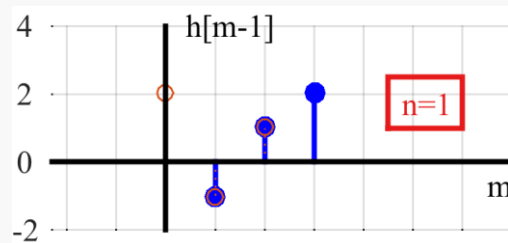
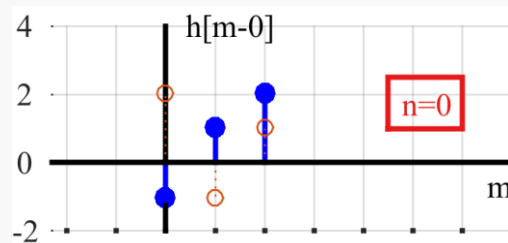
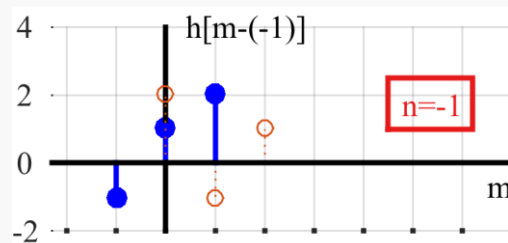
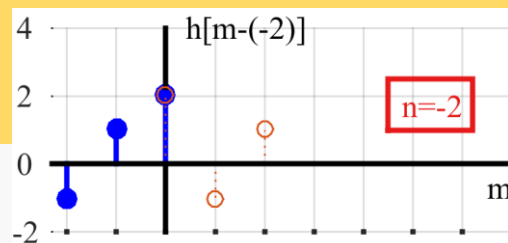
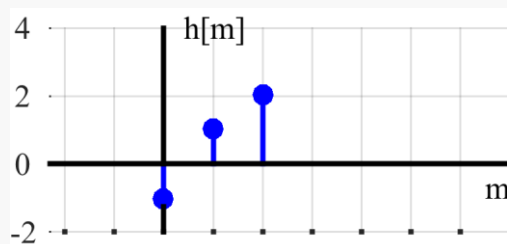
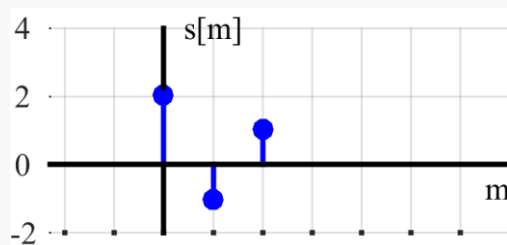
Correlation is the heart of a digital communication system

For two signals $s[n]$ and $h[n]$, correlation is defined as

$$\text{corr}[n] = \sum_{m=-\infty}^{+\infty} s[m]h^*[m-n]$$

1. Select the time shift n
2. Compute the sample-by-sample product $s[m]h^*[m-n]$ and add for all m
3. Repeat for all n

Example

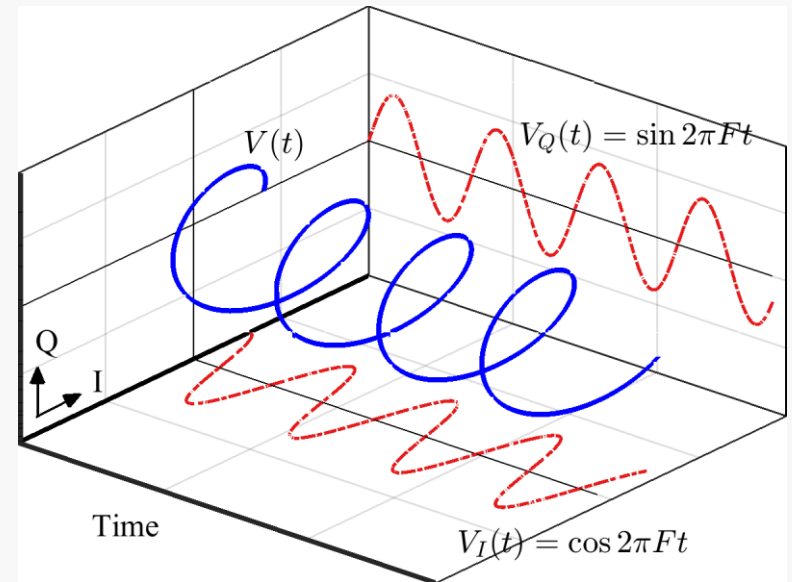


A Complex Sinusoid

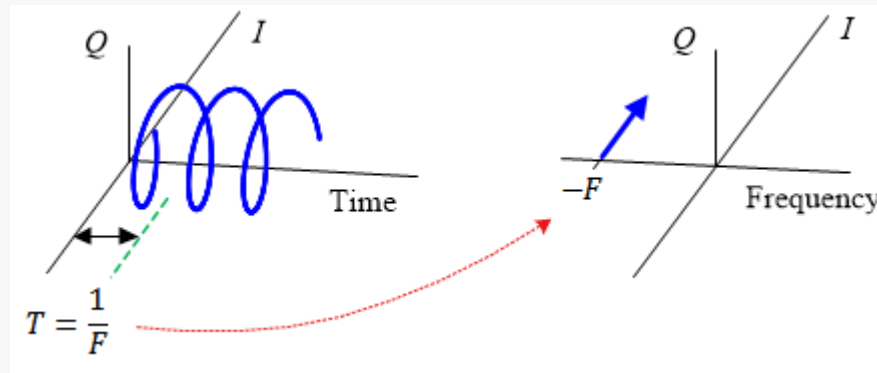
- Time domain: two real sinusoids

$$V_I(t) = \cos 2\pi Ft$$

$$V_Q(t) = \sin 2\pi Ft$$

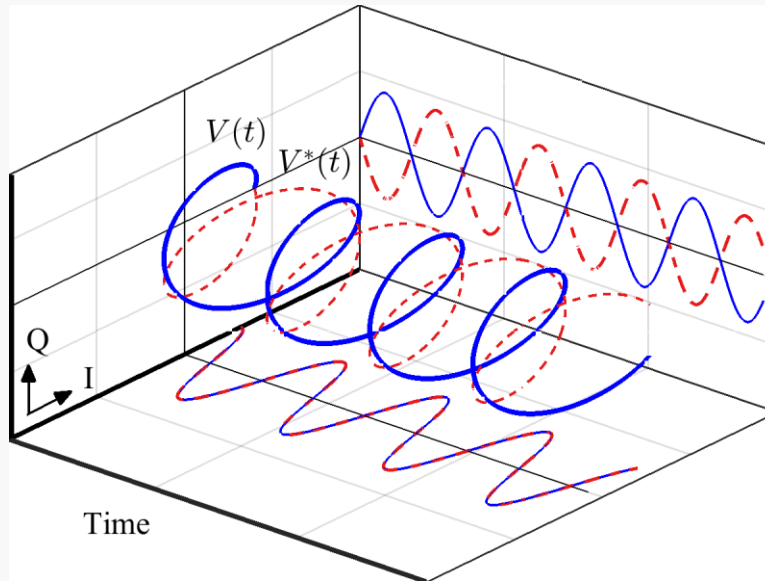


- Frequency domain: one 'tick' on the frequency axis

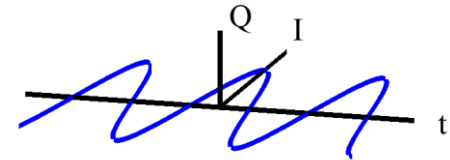


A Real Sinusoid

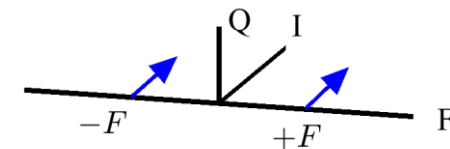
Interestingly, a real sinusoid is composed of two complex sinusoids



Cosine in Time IQ-Plane



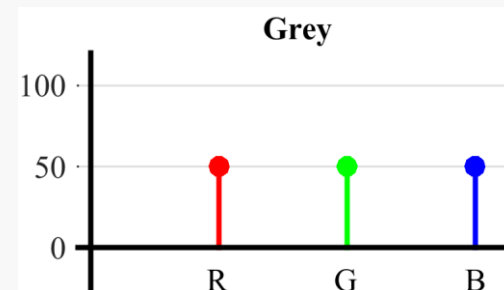
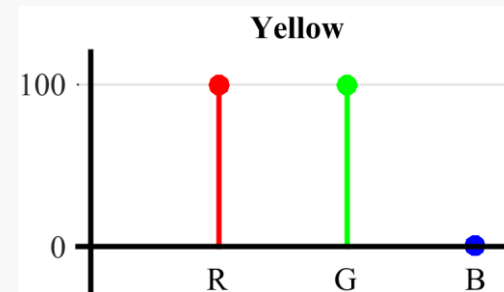
Cosine in Frequency IQ-Plane



Making Up a Colour

Red, Green, and Blue colours combine together to produce a broad range of different colours

- Yellow [1 1 0]
- Grey [0.5 0.5 0.5]

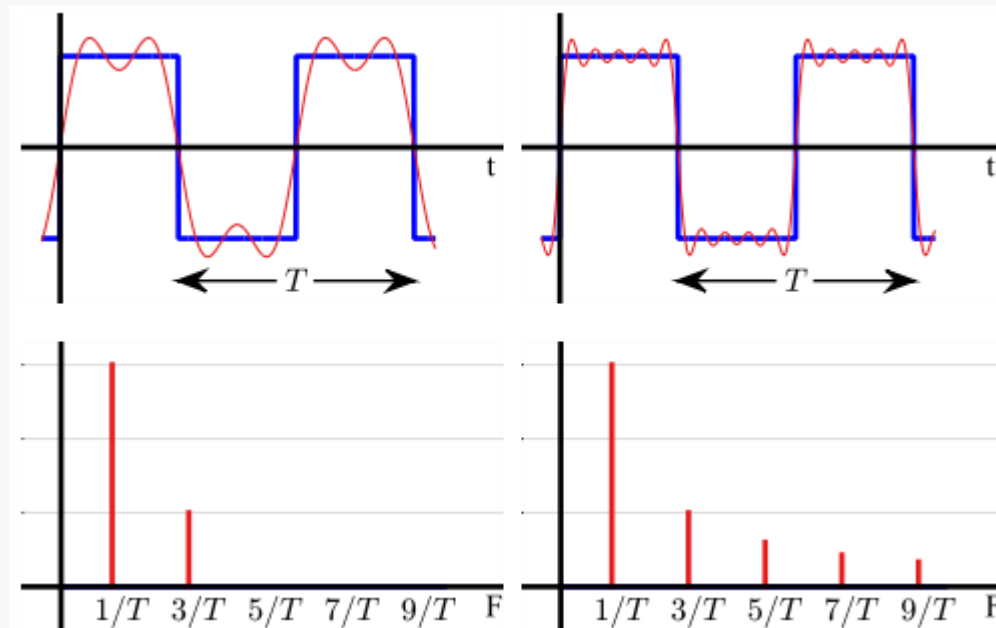


Making Up a Signal

Every signal is composed of sinusoids with different frequencies

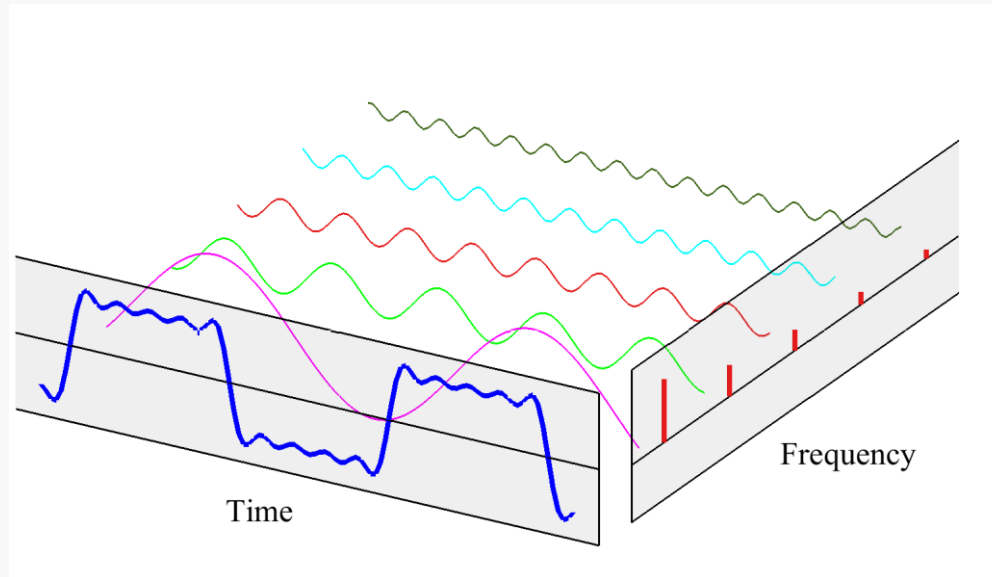
- A square wave is an example

A better approximation is achieved with more sinusoids



Square Wave

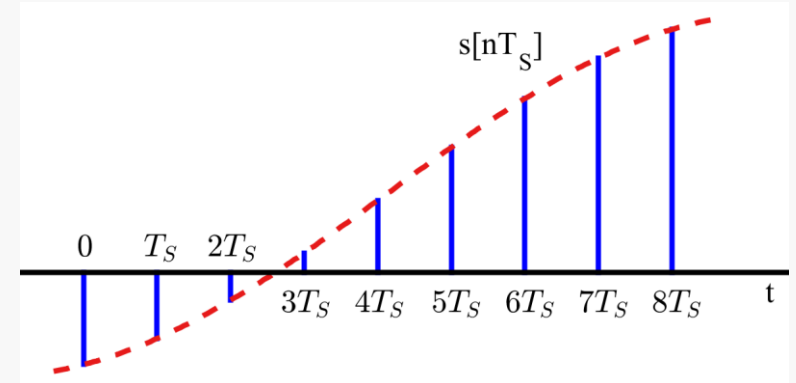
Time and frequency domains



Sampling

Samples at regular intervals T_S seconds apart

$$s[n] = s(t) \Big|_{t=nT_S}$$

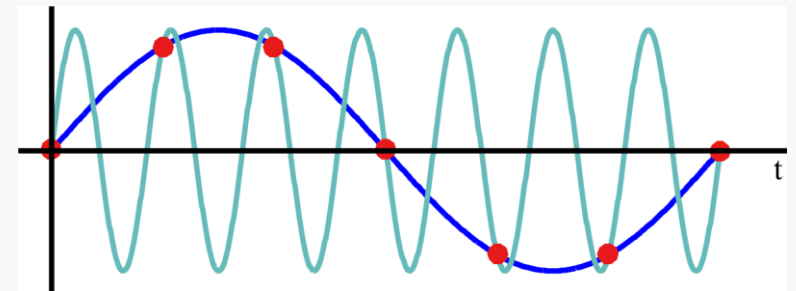


What happens to a continuous-time sinusoid?

$$\begin{aligned} s[n] &= s(nT_S) = A \cos(2\pi F n T_S + \theta) \\ &= A \cos\left(2\pi \frac{F}{F_S} n + \theta\right) \end{aligned}$$

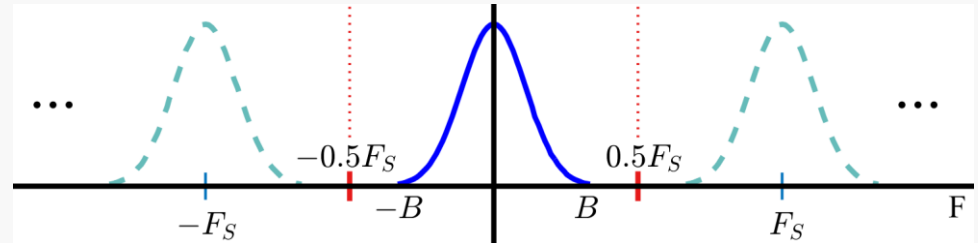
Remember

$$\cos \phi = \cos(2\pi + \phi)$$



Spectral Aliasing

- Aliases appear after sampling

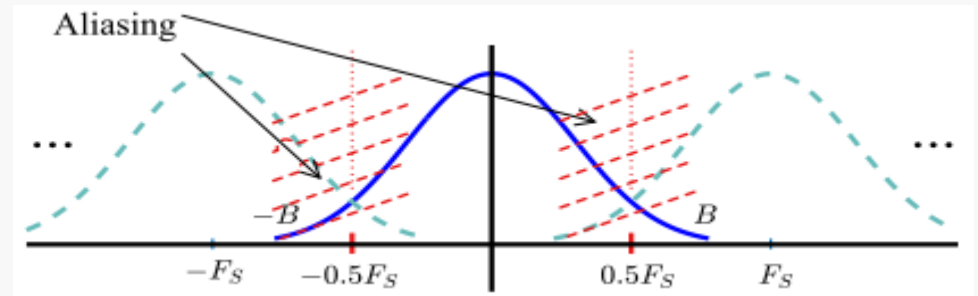


- The unique range of frequencies is

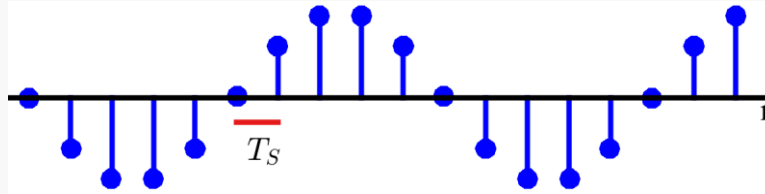
$$-0.5F_S \rightarrow +0.5F_S$$

- Sampling theorem:** To prevent distortion in the sampled signal

$$F_S > 2B$$



A Discrete-Time Sinusoid



- For $F_S = 10$ Hz

$$T = 10 \frac{\text{samples}}{\text{period}} \cdot 0.1 \frac{\text{seconds}}{\text{sample}} = 1 \text{ second}$$

- For $F_S = 500$ Hz

$$T = 10 \frac{\text{samples}}{\text{period}} \cdot 0.002 \frac{\text{seconds}}{\text{sample}} = 0.02 \text{ seconds}$$

Discrete Frequency

Collect N samples at a rate of $F_S = 1/T_S$ (total time is NT_S seconds)

- One full cycle in these N samples implies $1/NT_S$ Hz

$$V_I[n] = \cos 2\pi \frac{1}{NT_S} t \Big|_{t=nT_S} = \cos 2\pi \frac{1}{NT_S} nT_S = \cos 2\pi \frac{1}{N} n$$

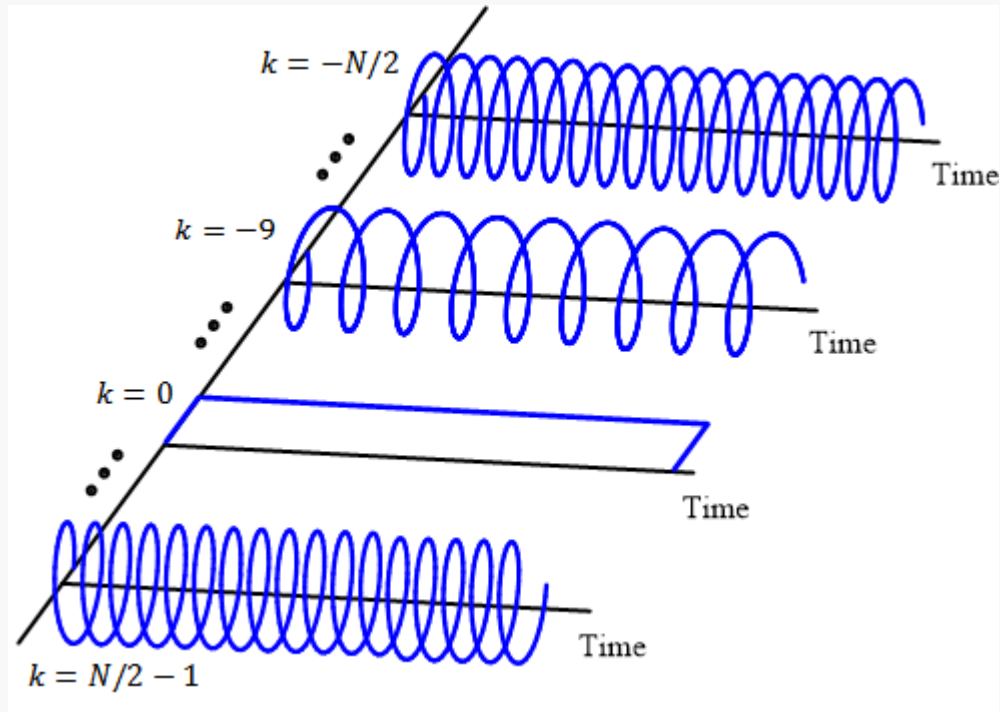
$$V_Q[n] = \sin 2\pi \frac{1}{NT_S} t \Big|_{t=nT_S} = \sin 2\pi \frac{1}{NT_S} nT_S = \sin 2\pi \frac{1}{N} n$$

- Total N complex sinusoids are possible with frequencies

$$0 \left(\frac{1}{N} \right), 1 \left(\frac{1}{N} \right), 2 \left(\frac{1}{N} \right), \dots$$

Set of Complex Sinusoids

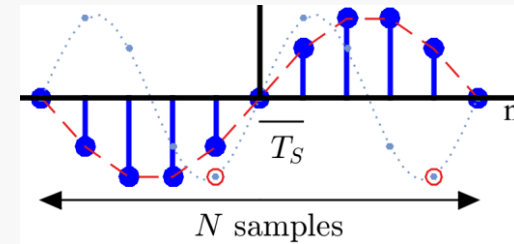
k cycles per NT_S seconds



Orthogonality

Orthogonality implies that their correlation at time shift zero is 0

$$\sum_{n=0}^{N-1} \sin 2\pi \frac{1}{N} n \cdot \sin 2\pi \frac{2}{N} n = 0$$

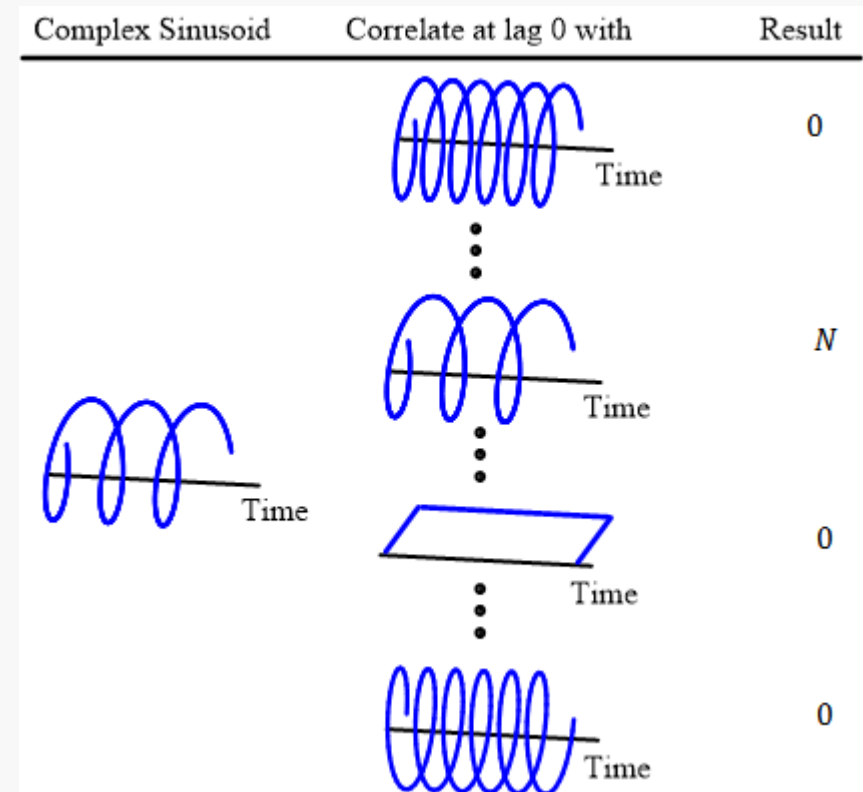


True for all integer multiplies of $1/N$

Orthogonality

For complex sinusoids

$$\sum_{n=0}^{N-1} \cos 2\pi \frac{k - k'}{N} n = \begin{cases} N & k = k' \\ 0 & k \neq k' \end{cases}$$
$$\sum_{n=0}^{N-1} \sin 2\pi \frac{k - k'}{N} n = 0$$



Discrete Frequency Axis

- The unique range is $-0.5F_S \rightarrow +0.5F_S$ or

$$-0.5 \leq \frac{F}{F_S} < +0.5$$

- It is divided into N distinct points, each of which is represented by

$$k = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, -1, 0, 1, \dots, \frac{N}{2} - 1$$

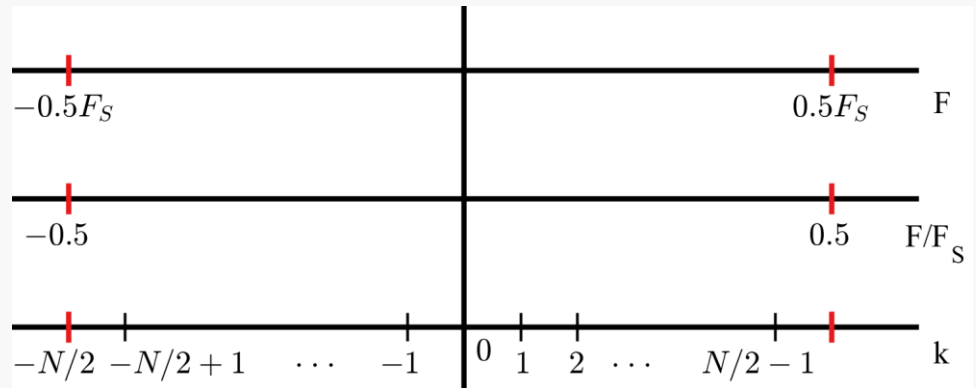
- Then, we have

$$\frac{k}{N} = -0.5, -0.5 + \frac{1}{N}, \dots, -\frac{1}{N}, 0, \frac{1}{N}, \dots, +0.5 - \frac{1}{N}$$

Discrete Frequency Axis

- Establishes relation between continuous and discrete frequencies

$$\frac{k}{N} = \frac{F}{F_S}$$

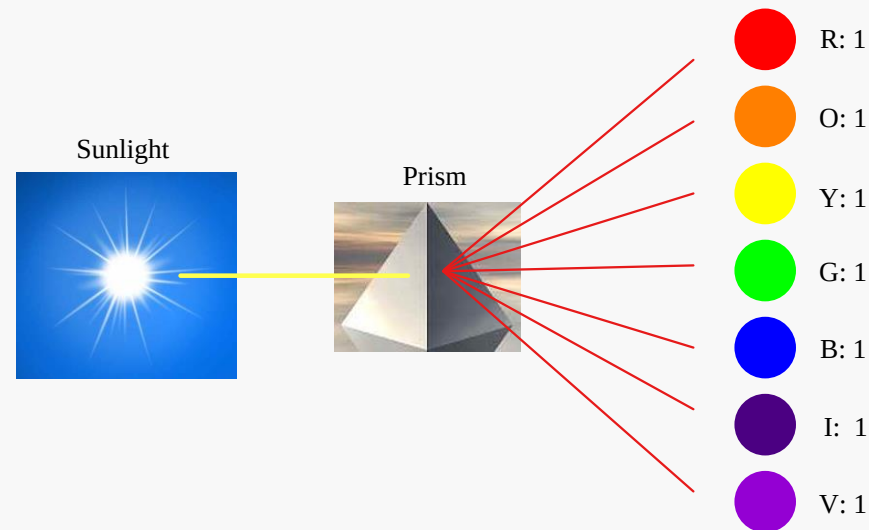


- With $F_S = 3$ kHz and $N = 32$

$$\begin{aligned} k = 0 & \rightarrow F = 3000 \cdot \frac{0}{32} = 0 \text{ Hz} \\ k = 1 & \rightarrow F = 3000 \cdot \frac{1}{32} = 93.75 \text{ Hz} \\ k = 2 & \rightarrow F = 3000 \cdot \frac{2}{32} = 187.5 \text{ Hz} \end{aligned}$$

White Light

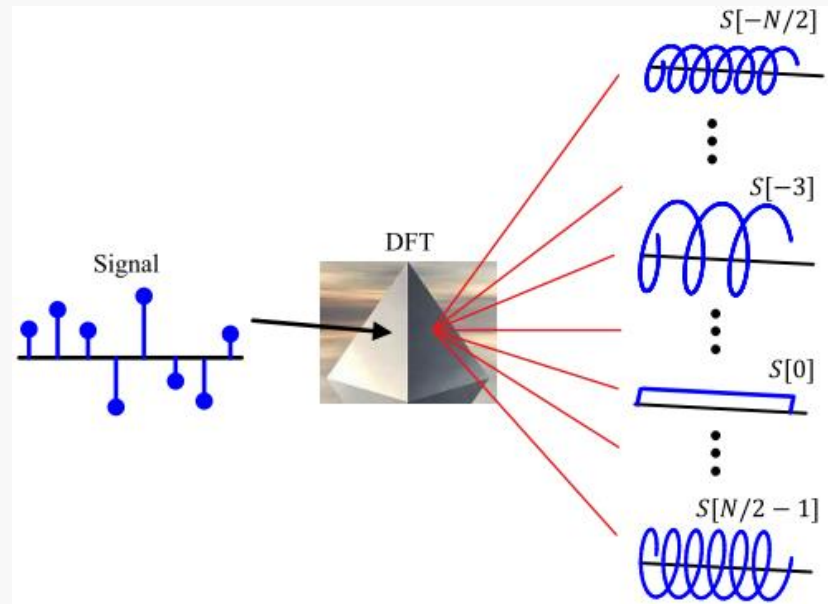
- In 1672, Newton demonstrated that the sunlight can be broken by a prism into 7 constituent colours of the rainbow
- A prism is a tool that tells us the intensity of each colour present in the white light



Discrete Fourier Transform (DFT)

- DFT finds **amplitude and phase** contributions in a signal from each of the N discrete-time complex sinusoids
- These reference sinusoids are called **analysis frequencies**

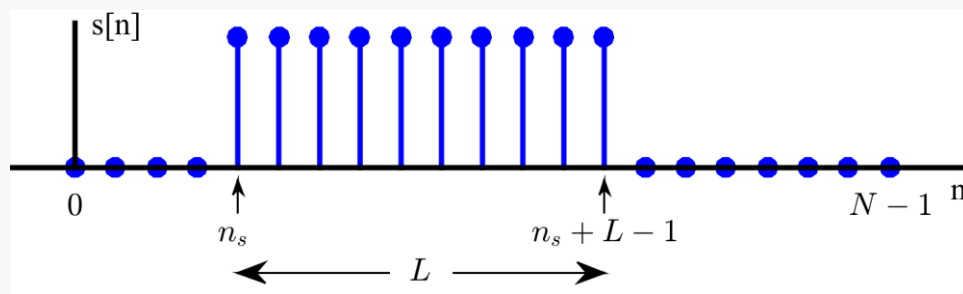
$$V_I[n] = \cos 2\pi \frac{k}{N} n$$
$$V_Q[n] = \sin 2\pi \frac{k}{N} n$$



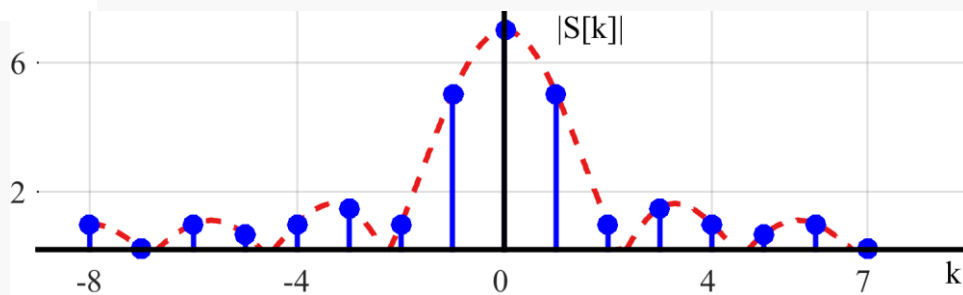
$$k = -\frac{N}{2}, -\frac{N}{2} + 1, \dots, -1, 0, 1, \dots, \frac{N}{2} - 1$$

Most Important DFT Pair

Rectangular signal



Sinc signal

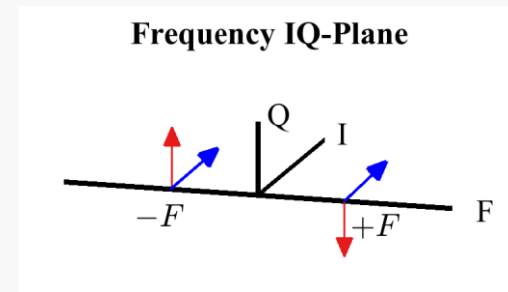
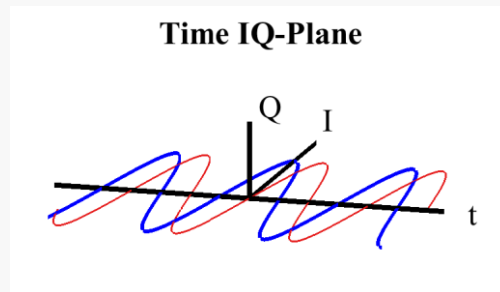


$$s[n] = \begin{cases} 1, & n_s \leq n \leq n_s + L - 1 \\ 0, & \text{otherwise} \end{cases}$$

$$|S[k]| = \frac{\sin \pi L k / N}{\sin \pi k / N}$$

Effect of Time Shift in Frequency Domain

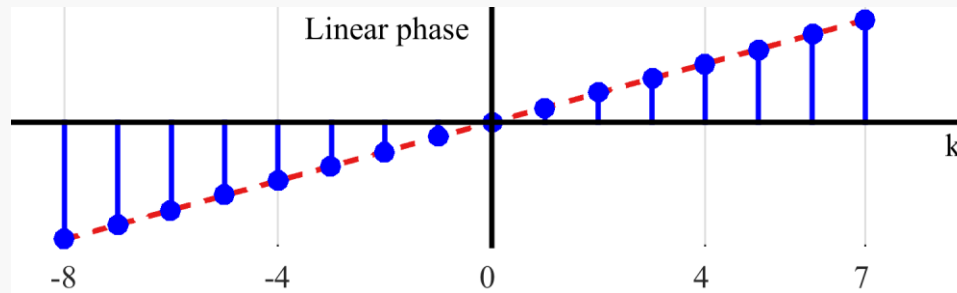
- Let us shift a cosine signal by $T/4$ to the right



$$\begin{aligned}\cos 2\pi F \left(t - \frac{T}{4} \right) &= \cos \left(2\pi F t - 2\pi F \frac{T}{4} \right) \\ &= \cos \left(2\pi F t - \frac{\pi}{2} \right) = \sin 2\pi F t\end{aligned}$$

Effect of Time Shift: Magnitude and Phase

Time shift $s[(n \pm n_0) \bmod N] \longrightarrow \pm 2\pi \frac{k}{N} n_0$ Phase shift

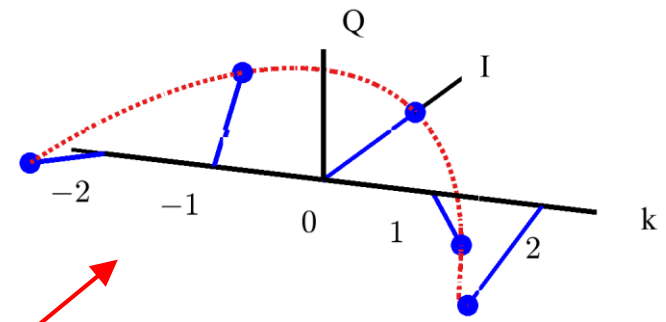
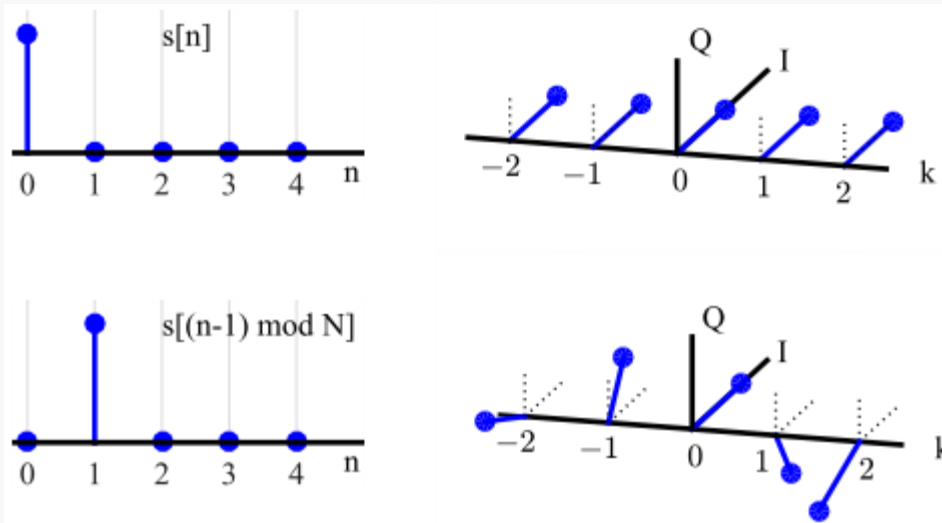


$$\begin{aligned} s(t) &\longrightarrow S(F) \\ s(t - t_0) &\longrightarrow S(F) \cdot \exp(-j2\pi F t_0) \end{aligned}$$

Effect of Time Shift: I and Q

Shift in one domain $\rightarrow$ Multiplication by a complex sinusoid in the other domain

A Frequency Domain Complex Sinusoid



A Digital Signal

- A discrete-time signal is sampled on time axis
- A digital signal is sampled at both time and amplitude axes

